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Started on Saturday, 31 August 2024, 9:41 PM

State Finished

Completed on Saturday, 31 August 2024, 10:26 PM

Time taken 44 mins 34 secs

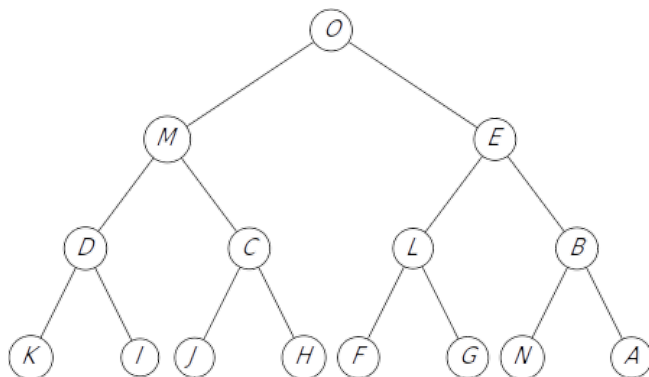
Grade 15.50 out of 28.00 (55%)

Question 1

Incorrect

Mark 0.00 out of 1.00

Consider the tree shown below:



How many nodes would have been first added to the Frontier and then popped from the frontier before reaching the goal state of K if Breadth First Tree Search is used?

- ☐ a. 5
- ☐ b. 6
- ☒ c. 7 ✖
- ☐ d. 3
- ☐ e. 4

Your answer is incorrect.

The correct answer is:

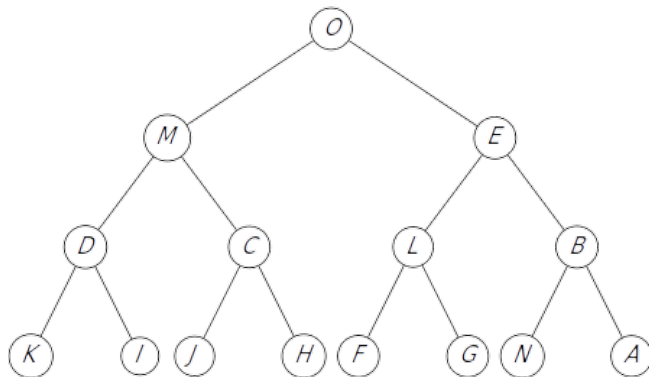
4

Question **2**

Incorrect

Mark 0.00 out of 1.00

Consider the tree shown below.



If DFS tree search is used to search for G in the tree, how many nodes would be added to the frontier before the goal is found.

- ☐ a. 13
- ☐ b. 7
- ☒ c. 11 ✖
- ☐ d. 14
- ☐ e. 9

Your answer is incorrect.

The correct answer is:

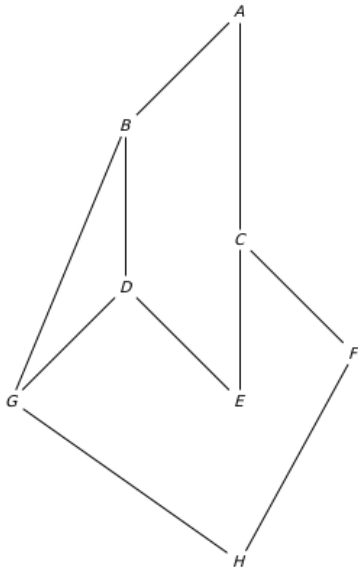
13

Question **3**

Correct

Mark 1.00 out of 1.00

Consider the graph below.



If Breadth First search on a graph is applied with the start node as A and the goal as H.

List the nodes in the order that they are placed into the frontier and then later expanded.

Assume that the children of each node are stored as below.

A: B C

B: G D

C: E F

D: G E

F: H

G: H

The graph is not directed -- If there is an edge from A to B then there is an edge from B to A

- ☐ a. A C B G D E F H
- ☐ b. A B C D E G F H
- ☐ c. A B C G D F E H
- ☒ d. A B C G D E F H ✓
- ☐ e. A B G D C E F H

Your answer is correct.

The correct answer is:

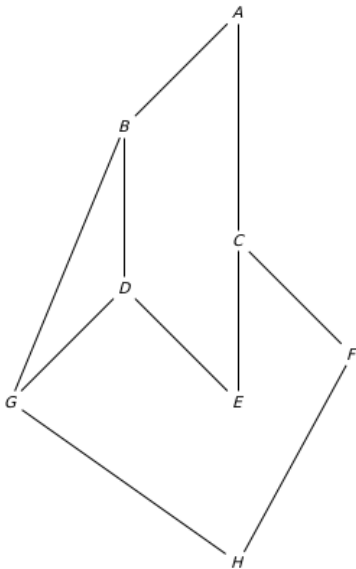
A B C G D E F H

Question **4**

Correct

Mark 1.00 out of 1.00

Consider the graph below.



If Breadth First search on a graph is applied with the start node as A and the goal as H.

List the nodes in the order that they are explored set.

Assume that the children of each node are stored as below.

A: B C

B: G D

C: E F

D: G E

F: H

G: H

The graph is not directed -- If there is an edge from A to B then there is an edge from B to A

Answer:



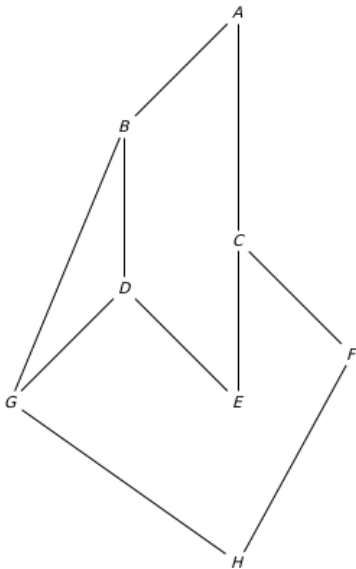
The correct answer is: A B C G D E F H

Question **5**

Correct

Mark 1.00 out of 1.00

Consider the graph below.



If Depth First Search on a graph is applied with the start node as A and the goal as H.

List the nodes in the order that they are placed into the frontier and then later expanded.

Assume that the children of each node are stored as below.

A: B C

B: G D

C: E F

D: G E

F: H

G: H

The graph is not directed -- If there is an edge from A to B then there is an edge from B to A

- ☐ a. A C E D G H
- ☐ b. A B D E C F H
- ☒ c. A B G H ✓
- ☐ d. A B D G H
- ☐ e. A C F H

Your answer is correct.

The correct answer is:

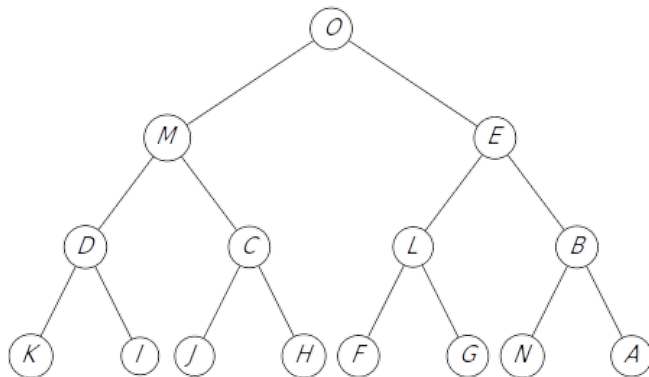
A B G H

Question **6**

Correct

Mark 1.00 out of 1.00

Consider the tree shown below:



Depth First Search is normally presented as a recursive algorithm.

How many recursive calls would be made (including the call to the root of the tree) to find the goal state, K, if a recursive Depth First Search was used?

- ☐ a. 5
- ☒ b. 4 ✓
- ☐ c. 3
- ☐ d. 6
- ☐ e. 2

Your answer is correct.

Frontier is O

Explore O – frontier of M, E --- 1 call

explore M – frontier is D, C, E --- 2 calls

explore D – frontier is K, I, C, E --- 3 calls

explore K – 4 calls..... Goal state is found.

The correct answer is:

4

Question **7**

Correct

Mark 1.00 out of 1.00

Is depth first search optimal?

Answer: No



No... It finds the "leftmost" solution if the search is done left to right (the children are specified in a left to right order).

The correct answer is: no

Question **8**

Not answered

Marked out of 1.00

Consider the depth-limited search algorithm (DLS), which executes depth-first search to a fixed depth D . Select the statements about DLS below that are true:

- ☐ a. DLS is complete and optimal
- ☐ b. DLS is complete, but not optimal
- ☐ c. DLS is neither complete nor optimal
- ☐ d. DLS is optimal, but not complete

Your answer is incorrect.

The correct answer is:

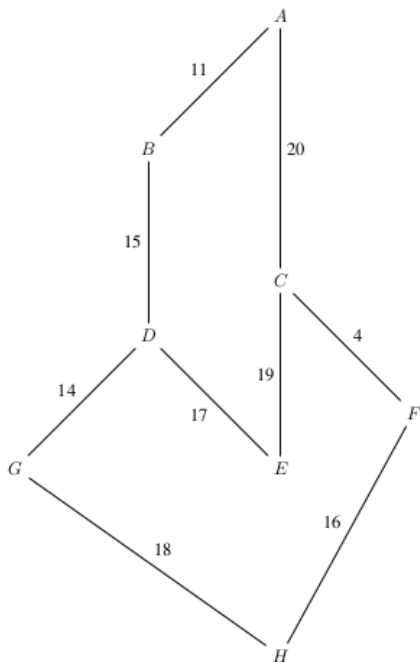
DLS is neither complete nor optimal

Question 9

Partially correct

Mark 1.50 out of 2.00

Consider the graph shown below.



Perform Uniform Cost Search for a path from A to H.

The shortest path is A C F H and the cost is 40.

How many nodes are added to the frontier and then expanded as this path is computed?

- ☐ a. 4
- ☐ b. 5
- ☐ c. 8
- ☐ d. 6
- ☒ e. 7 ✓

Your answer is partially correct.

The correct answer is:

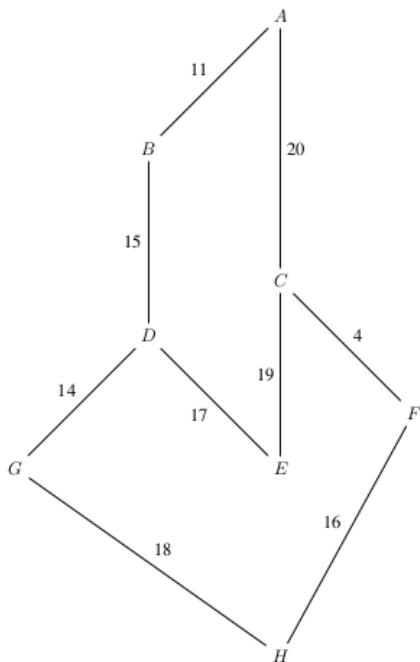
8

Question 10

Incorrect

Mark 0.00 out of 1.00

Consider the graph shown below.



If Uniform Cost search is applied to this graph, is any node considered for addition to the frontier but not added as there is already a shorter path under consideration through that node?

Answer with

yes, node (for example, "yes, Z")

or

no

Answer: yes, B



The correct answer is: yes, E

Question **11**

Correct

Mark 1.00 out of 1.00

Assume that we are trying to determine the optimal plan to solve a problem with uniform costs on a computer with limited memory. What search method is most suitable in this instance?

- ☐ a. Breadth-first search
- ☐ b. Uniform-cost search
- ☒ c. Iterative deepening search ✓
- ☐ d. Depth-first search

Your answer is correct.

The correct answer is:

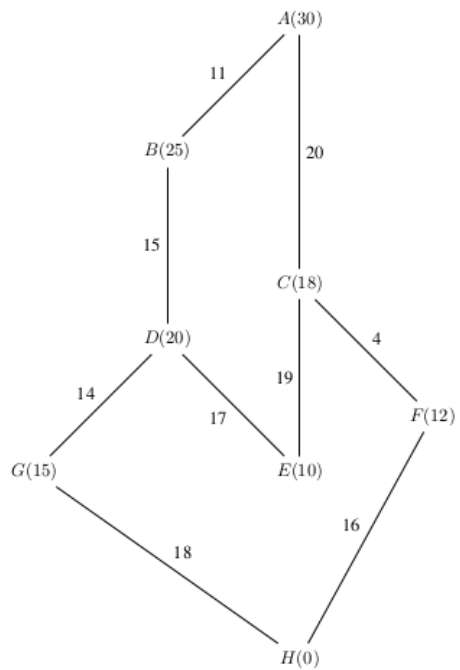
Iterative deepening search

Question 12

Incorrect

Mark 0.00 out of 1.00

Consider the graph shown below.



Here the number in brackets after the node name is the heuristic for that node.

If Step 1 is expanding the start node (A), what nodes will be in the frontier (in the correct order) after 3 steps?

Give your answer in the form of X,Y,Z (node,node,node,node --- i.e comma separators and no spaces.

Answer: A,C,D,G,H



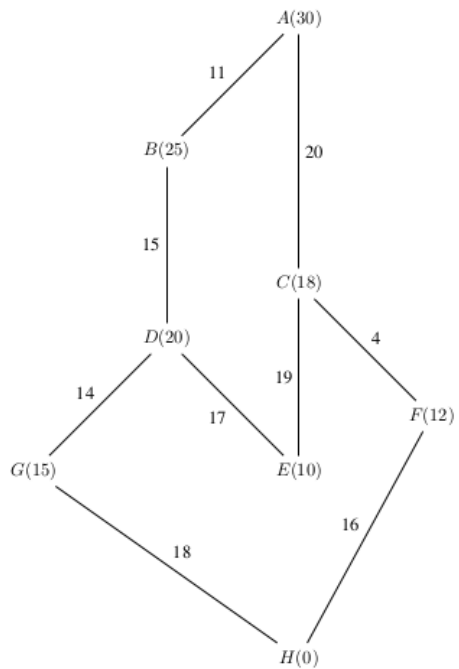
The correct answer is: F,D,E

Question 13

Not answered

Marked out of 1.00

Consider the graph shown below.



Here the number in brackets after the node name is the heuristic for that node.

Step 1 is expanding the start node (A)

What will be the smallest value ($g(n) + h(n)$) in the frontier after step 4?

Answer:



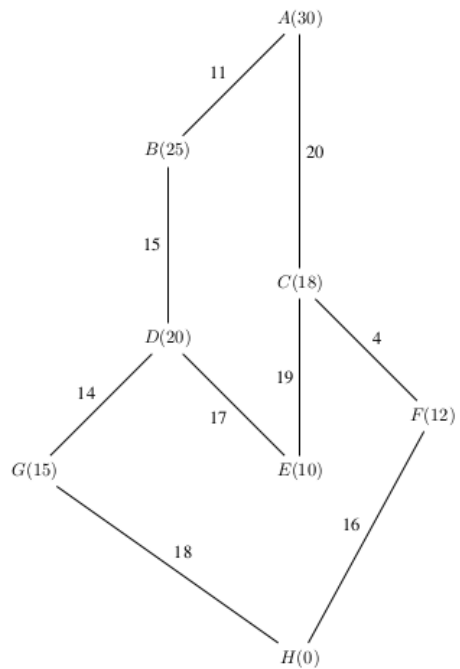
The correct answer is: 40

Question 14

Not answered

Marked out of 1.00

Consider the graph shown below.



Here the number in brackets after the node name is the heuristic for that node.

Suppose the heuristic at node C was made 100 instead of 18.

What would the effect on the solution be?

- ☐ a. The same solution would be found in the same number of steps.
- ☐ b. No solution would be found.
- ☐ c. A different path with a different value would be found
- ☐ d. A longer path through different nodes would be found
- ☐ e. The same solution would be found but it would require more steps

Your answer is incorrect.

The correct answer is:

A longer path through different nodes would be found

Question 15

Correct

Mark 1.00 out of 1.00

Let $g(n)$ be the total cost so far from the root node to node n , and let $h(n)$ be the heuristic estimate from node n to the goal. Select the statements below that are true

- ☐ a. If we implement A* search with $h(n) = -g(n)$, then our algorithm reduces to greedy best first search
- ☒ b. If we implement A* search with $h(n) = 0$, then our algorithm reduces to uniform cost search ✓
- ☐ c. If we implement A* search with $h(n) = 0$, then our algorithm reduces to breadth first search
- ☐ d. If we implement A* search with $h(n) = 0$, then our algorithm reduces to greedy best first search

Your answer is correct.

The correct answer is:

If we implement A* search with $h(n) = 0$, then our algorithm reduces to uniform cost search

Question 16

Incorrect

Mark 0.00 out of 1.00

Let $g(n)$ be the total cost from the start to node n , and let $h(n)$ be a consistent heuristic estimate of node n . Assume that we are executing A* search with consistent heuristic h , and that the cost of the optimal path to reach the goal from the start node is C . Select the statement below that is true:

- ☒ a. All nodes n that such that $g(n) < C$ will be expanded during the search ✗
- ☐ b. All nodes n that such that $g(n) + h(n) < C$ will be expanded during the search
- ☐ c. All nodes n that such that $h(n) < C$ will be expanded during the search
- ☐ d. We do not have enough information to make a claim about which nodes will be expanded

Your answer is incorrect.

The correct answer is:

All nodes n that such that $g(n) + h(n) < C$ will be expanded during the search

Question **17**

Partially correct

Mark 4.00 out of 5.00

Match the following statements.

A heuristic is admissable if the

heuristic, $h(n)$, is less than or equal to the True Future Cost of finding the goal from node n .



The space complexity of Depth First Search is

proportional to the branching factor multiplied by the maximum depth of any node.



An agent is

a decision making entity



The time complexity of Breadth First Search is

proportional to the branching factor multiplied by the depth of the shallowest solution.



The state space is

the set of all possible states that could occur in some game or planning project.



Your answer is partially correct.

You have correctly selected 4.

The correct answer is:

A heuristic is admissable if the → heuristic, $h(n)$, is less than or equal to the True Future Cost of finding the goal from node n .

The space complexity of Depth First Search is → proportional to the branching factor multiplied by the maximum depth of any node.

An agent is → a decision making entity,

The time complexity of Breadth First Search is → exponential in terms of the depth of the search space.

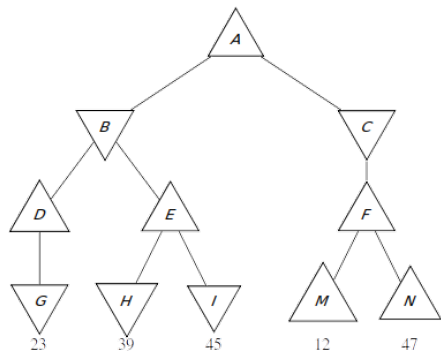
The state space is → the set of all possible states that could occur in some game or planning project.

Question **18**

Incorrect

Mark 0.00 out of 1.00

Consider the game tree shown below:



What utility value will be assigned to node *B* by the minimax algorithm?

Answer: ❌

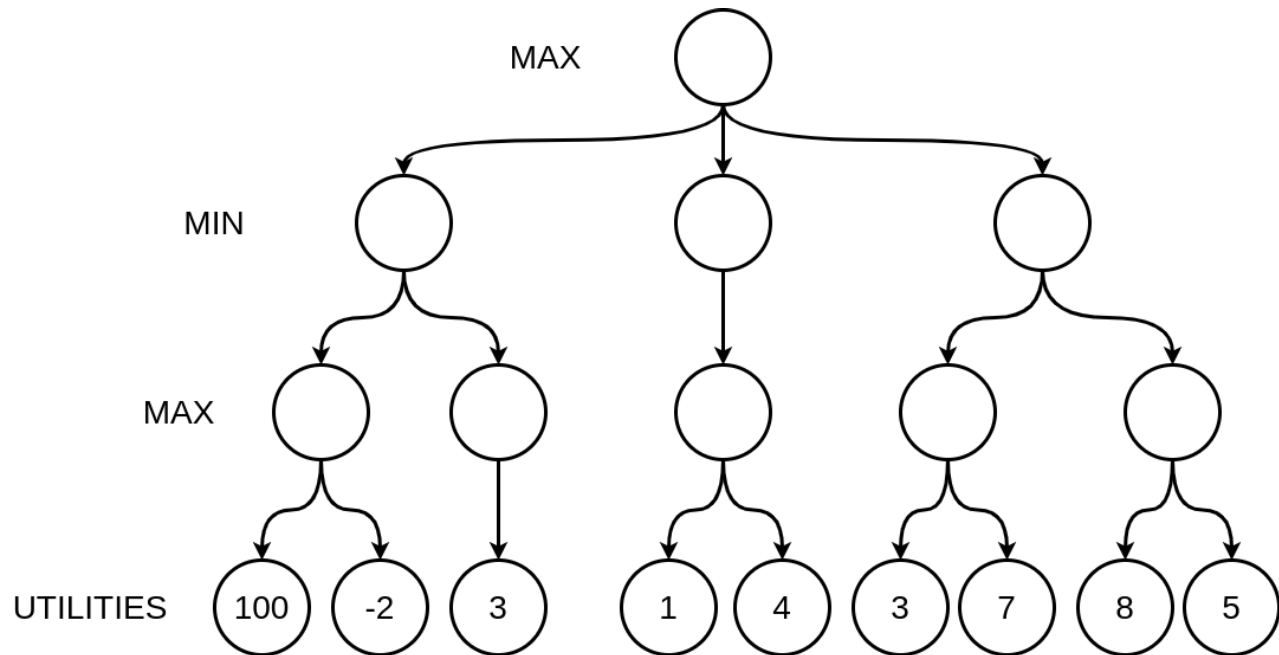
The correct answer is: 23

Question 19

Incorrect

Mark 0.00 out of 1.00

Consider a two player competitive game played by a minimising and maximising player. Given the game tree below, what is the value at the root node according to minimax?



Answer:

1

✖

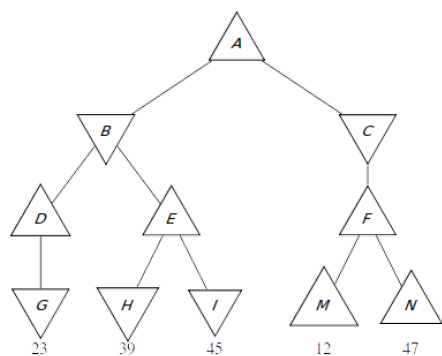
The correct answer is: 7

Question **20**

Correct

Mark 2.00 out of 2.00

Consider the game tree shown below:



Suppose that alpha-beta pruning is applied to the tree in a left to right fashion.

Which of the following statements is true?

- ☐ a. Nodes I and N are pruned.
- ☒ b. Node I is pruned. ✓
- ☐ c. Node F is pruned
- ☐ d. Node N is pruned
- ☐ e. There is no pruning

Your answer is correct.

The correct answer is:

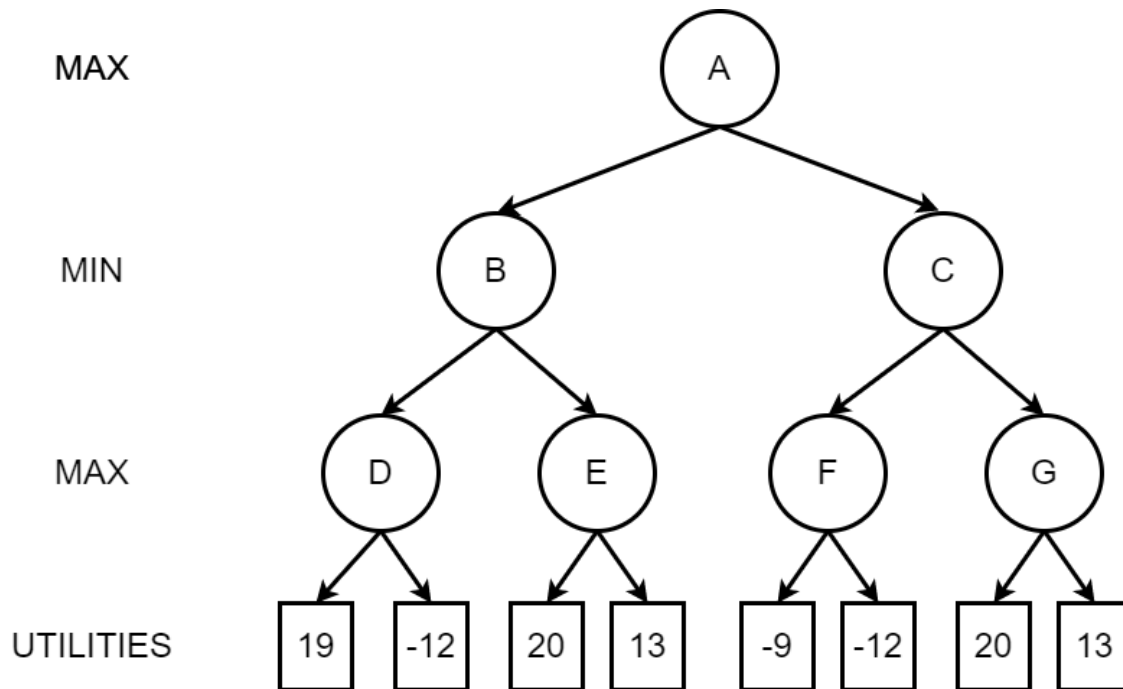
Node I is pruned.

Question 21

Partially correct

Mark 1.00 out of 2.00

Consider a two-player competitive game played by a minimising and maximising player. Assume that we apply the Alpha-beta pruning algorithm on the game tree below, where nodes are labelled by letters for ease of reference. The depth at which the maximising and minimising players make decisions is also indicated. Further, assume that children are considered from left to right. Select the statements below that are true:



- ☒ a. The right child of node C will be pruned, as will the right child of node E. ✓
- ☐ b. The right child of node B will be pruned, as will the right child of node F.
- ☒ c. The value at the root node after the algorithm completes is 13. ✗
- ☐ d. The value at the root node after the algorithm completes is -9.
- ☐ e. The value at the root node after the algorithm completes is -12.
- ☐ f. The right child of node A will be pruned.
- ☐ g. The left child of node C will be pruned, as will the left child of node E.
- ☐ h. The value at the root node after the algorithm completes is 19.
- ☐ i. The value at the root node after the algorithm completes is 20.
- ☒ j. No pruning of nodes or edges takes place. ✗

Your answer is partially correct.

You have selected too many options.

The correct answers are:

The right child of node C will be pruned, as will the right child of node E,

The value at the root node after the algorithm completes is 19.

Jump to...

[Quiz 3 Classical Search and Sorting ►](#)